Session 9: Multiple Linear Regression

Context & Objective

Textbook: Chapters 15 and 16 – Multiple Linear Regression & Model Selection

Objective: Students build a multiple regression model to estimate housing listing prices based on square footage, location dummy variables, and property age. Students learn to interpret coefficients, assess overall model significance, and audit for multicollinearity using Variance Inflation Factors (VIF).

1. Antigravity Skill Definition (multiple-regression-solver)

Skill Name: multiple-regression-solver

Purpose: Evaluates multiple linear regression models with 2 or more independent variables. Calculates regression coefficients, assesses model fit (R^2, adjusted R^2, F-statistic), tests individual predictor significance, and diagnoses multicollinearity through Variance Inflation Factors.

1.1 Core Logic & Formulas

Step 1: Input Dataset with Y and Multiple X Variables

Y = Dependent variable (response) – housing price in thousands of dollars

X_1 = Size in square feet (continuous predictor)

X_2 = Location dummy variable (1 = city center, 0 = suburb)

X_3 = Age in years (continuous predictor)

Step 2: Estimate Regression Coefficients (Ordinary Least Squares)

Multiple Regression Model: $\hat{y} = b_0 + b_1X_1 + b_2X_2 + b_3X_3$

b_0 = Intercept (baseline price)

b_1 = Effect of size on price (controlling for location and age)

b_2 = Premium for city center location (controlling for size and age)

b_3 = Depreciation per year of age (controlling for size and location)

Step 3: Assess Overall Model Fit

R^2 = SSR / SST – proportion of price variation explained by all predictors

$R^2_adj = 1 - (1 - R^2) \times (n-1) / (n-p-1)$ – penalizes adding non-informative predictors

F-statistic = (SSR/p) / (SSE/(n-p-1)) – tests if model is jointly significant

Step 4: Test Individual Predictor Significance

For each coefficient b_j: t-statistic = b_j / SE(b_j) with df = n - p - 1

H_0: $\beta_j = 0$ (predictor has no effect controlling for others) vs H_1: $\beta_j \neq 0$

If p-value < α: Reject H_0 (predictor significantly impacts Y)

Step 5: Audit Multicollinearity Using VIF

Problem: High correlation among predictors inflates standard errors and destabilizes estimates

$VIF_j = 1 / (1 - R^2_j)$ where R^2_j is from regressing X_j against all other X variables

Interpretation:

VIF < 5: Acceptable (minimal multicollinearity)

5 ≤ VIF < 10: Moderate multicollinearity (investigate)

VIF ≥ 10: Severe multicollinearity (remove or combine predictors)

2. Student Execution Prompt

Evaluate factors influencing home price (Y in $1000s) based on:

X_1: Size in square feet (continuous)
X_2: Location dummy variable (1 = city center, 0 = suburb)
X_3: Property age in years (continuous)

Regression Model Output Provided:

Coefficients: $b_0 = 50.0$, $b_1 = 0.12$ ($p = 0.01$), $b_2 = 45.0$ ($p = 0.03$), $b_3 = -1.5$ ($p = 0.02$)
Overall Model $F = 22.4$ ($p < 0.001$), Adjusted $R^2 = 81.2\%$
VIF factors: $VIF_1 = 1.8$, $VIF_2 = 3.2$, $VIF_3 = 2.4$

Task: Interpret each coefficient in plain business language. Audit the model for multicollinearity. Predict the price for a 2,000 sq ft city center home built 10 years ago.

Regression Coefficients & Significance Tests:

Variable	Coefficient	Interpretation	p-Value
Intercept (b_0)	$50k	Baseline price	—
Size (b_1)	$0.12/sq ft	Each sq ft adds $120 to price	0.01 **
Location (b_2)	$45k	City center premium vs suburb	0.03 *
Age (b_3)	-$1.5k/yr	Each year depreciates price by $1,500	0.02 *

Multicollinearity Audit (Variance Inflation Factors):

Variable	VIF	Severity	Status
Size (X_1)	1.8	Low	✓ OK
Location (X_2)	3.2	Low	✓ OK
Age (X_3)	2.4	Low	✓ OK

3. Managerial Decision Output

Model Summary & Interpretation:

Overall Model Fit: $F(3, n-4) = 22.4$, $p < 0.001$, Adjusted $R^2 = 81.2\%$
All three predictors are individually statistically significant ($p < 0.05$)
No multicollinearity concerns: all VIF values < 5 (range: 1.8 to 3.2)

Coefficient Interpretation:

Each additional square foot adds $120 to a home's price (controlling for location and age)
City center homes command a $45,000 premium over identical suburban properties
Each year of age depreciates the home by $1,500 (controlling for size and location)

Pricing Example & Decision:

For a 2,000 sq ft city center home built 10 years ago:

$$\hat{y} = 50.0 + 0.12(2{,}000) + 45.0(1) - 1.5(10)$$
$$\hat{y} = 50.0 + 240.0 + 45.0 - 15.0 = \$320{,}000$$

The real estate analyst concludes that the regression model is robust with excellent predictive power (adjusted $R^2 = 81.2\%$) and no multicollinearity issues. All coefficients are statistically significant and carry meaningful business interpretations. The model predicts a fair market listing price of $320,000 for a 2,000 sq ft city center home aged 10 years. The analyst will use this model to set competitive listing prices across their portfolio, adjusting for size, location, and property age. The strong statistical significance and absence of collinearity provides confidence that these pricing recommendations rest on solid empirical foundations.

